

## PART A – OBJECTIVE QUESTIONS

[ 20 marks ]

Part A consists of 20 objective questions. Each question carries 1 mark.

Choose the correct answer and write your answer in the test booklet.

### CH 4A Search Algorithms (3/5)

1. What is the evaluation function in A\* approach?

- a) Heuristic function
- b) Path cost from start node to current node
- c) Path cost from start node to current node + Heuristic cost
- ☒ d) Average of Path cost from start node to current node and Heuristic cost

2. Heuristic function  $h(n)$  is \_\_\_\_\_

- a) Lowest path cost
- b) Cheapest path from root to goal node
- c) Estimated cost of cheapest path from root to goal node
- ☒ d) Average path cost

3. A heuristic is a way of trying \_\_\_\_\_

- a) To discover something or an idea embedded in a program
- b) To search and measure how far a node in a search tree seems to be from a goal
- c) To compare two nodes in a search tree to see if one is better than another
- ☒ d) All of the mentioned

4. A\* algorithm is based on \_\_\_\_\_

- a) Breadth-First-Search
- b) Depth-First –Search
- ☒ c) Best-First-Search

d) Hill climbing

5. Best-First search is a type of informed search, which uses \_\_\_\_\_ to choose the best next node for expansion.

a) Evaluation function returning lowest evaluation

b) Evaluation function returning highest evaluation

c) Evaluation function returning lowest & highest evaluation

d) None of them is applicable

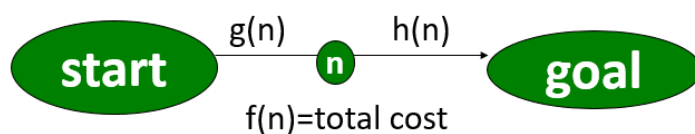
#### CH 4B Search Algorithms -HEURISTIC EVALUATION (2/4)

1. Which evaluation criterias can be applied on a heuristic function?

- I. Informedness
- II. Admissibility
- III. Monotonocity
- IV. Consistency

- A. I and IV
- B. I, II, and III
- C. I, II, III and IV
- D. I and III

2. Which statements are TRUE about the Figure 1 below



- I.  $f(n) = g(n) + h(n)$  estimates the total cost of path from start state through n to the goal state
- II.  $g(n)$  is the actual distance from n to start state
- III. For 8 puzzle problem,  $h(n)$  could be no of tiles in wrong position
- IV.  $g(n)$  is the actual distance from n to goal state

- A. I and IV
- B. I, II, and III
- C. I, II, III and IV
- D. I and III

3. For a consistent heuristic function to be consistent, it must fulfill the condition  $h(n) \leq \text{cost}(n, n') + h(n')$  which means the \_\_\_\_\_ cost of node n to the goal must be equal or less than the \_\_\_\_\_ cost of node n to its successor node plus the \_\_\_\_\_ cost of the successor node to the goal.

- A. estimated, true, estimated
- B. true, estimated, true
- C. heuristic, true, heuristic
- D. shortest, estimated, shortest

4. Which statements are TRUE about A\* algorithm

- I. A\* algorithm is optimal if its  $h(n)$  is admissible
- II. All A\* algorithms are admissible
- III. Monotone or consistent heuristic is an admissible A\* algorithm
- IV. Not all admissible A\* algorithms are monotone or consistent

- A. I and IV
- B. I, II, and III
- C. I, II, III and IV
- D. I and III

#### **CH 4C Problem Solving with search – (6/8)**

1. Alpha Beta pruning algorithm follows \_\_\_\_\_ search and allows pruning of branches in \_\_\_\_\_ fashion.

- A. Breadth-first, backtracking
- B. Depth-first, backtracking
- C. Best-first, forward chaining
- D. Best-first, backtracking

2. Which of the following statement is TRUE about Alpha Beta procedure?

- A. Alpha Beta pruning algorithm follows best-first search strategy.

B. *n-ply-look-ahead* can be applied in a complicated games where it is impossible to expand the state graph out to leaf nodes.

C. In alpha beta pruning, Max will update  $\beta$  value whilst Min will update  $\alpha$  value.

D. In alpha beta procedure, two values are created, i.e  $\alpha$  associates with MAX-never decrease and  $\beta$  associates with MIN-never increase

3. Which of the following statement is FALSE about Alpha Beta rules?

A. While backtracking, the node values are passed to the parent

B. While backtracking, the  $\alpha$  and  $\beta$  values are passed to the parent

C. MAX only update  $\alpha$ , MIN only update  $\beta$

D. Condition to fulfill when pruning is :  $\alpha \geq \beta$

4. Which of the following statement is FALSE about Alpha Beta procedure?

A. It provides faster searching because it can prune a leaf or entire branch at any depth of tree.

B. It improves search efficiency in two-person games compared to minimax that always pursues all branches in state space

C. Alpha ( $\alpha$ ) is the highest value found along the path of MAX and start at  $\infty$

D. Alpha ( $\alpha$ ) is the highest value found along the path of MAX and start at  $-\infty$

### MINIMAX

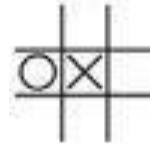
2. According to the rule of MINIMAX, if the parent state is at minimum, the \_\_\_\_\_ value from its children will be assigned to the parent.

- A. minimum
- B. maximum
- C. 0
- D. 1

3. A leaf node will be labelled as \_\_\_\_\_ if the turn to play is MIN and \_\_\_\_\_ if the turn to play is MAX.

- A. 1, 0
- B. 0, 1**
- C. minimum value from the child, maximum value from the child
- D. maximum value from the child, minimum value from the child

4. What is the heuristic value of the following tic-tac-toe state if the position of X and O is as follows?



6-4

- A. 2
- B. 4**
- C. 6
- D. -1

1. Which of the following statements about recursion -based system is FALSE?

- A. Recursive step-procedure calls itself
- B. It has a termination condition that stops the procedure from recurring infinitely
- C. Natural control for functions and procedures
- D. Useful in searching algorithm**

2. The components in production systems includes the following

- I. Production Rules
- II. Working memory
- III. Recognize-act control cycle
- IV. Conflict Routine

- A. I, II, III**
- B. I, II, IV
- C. II, III, IV
- D. I, III, IV

3. \_\_\_\_\_ is an inference technique that uses IF-THEN rules to repetitively breaks a goal into smaller sub-goals that are easier to prove.

- A. Forward chaining
- B. Backward chaining
- C. Heuristic search**
- D. Exhaustive search

1. AI Agent use \_\_\_\_\_ and \_\_\_\_\_ to perceive and act based on the environment.

- A. Sensors, Actuators
- B. Sensors, Performance**
- C. Sensors, Action
- D. Sensors, Goal

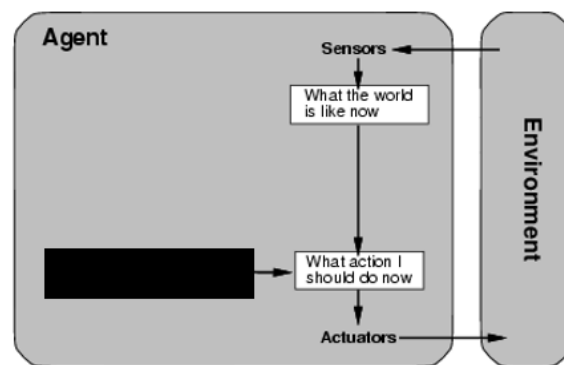


Figure 1: AI Agent

2. What is the suitable rule to be implemented by an AI agent in Figure 1?

- A. Simple action rule
- B. Condition action rule**
- C. Performance action rule
- D. Goal action rule

3. What is the type of intelligent agent which relies on current conditions without referring to the historical data?

- A. Utility-based
- B. Goal-based
- C. Simple reflex**
- D. Learning

Answer: C

4. Which is best to describe the purpose of an intelligent agent?
- A. To respond based on the surrounding environments.
  - B. To facilitate human-to-computer interaction.
  - C. To determine whether an answer is right or wrong.
  - D. To acquire vast amounts of searchable data.

5. An agent must perceive data using \_\_\_\_\_ in order to operate in the environment.
- A. Actuators
  - B. Sensors
  - C. Factors
  - D. Refactors

Answer: B

1. Which one of the following is **TRUE** about clustering?

- I. It follows trial and error fashion
- II. It uses similarity measures to group the data.
- III. It is a reward-based technique
- IV. It learns from unlabelled data

- A. I and IV
- B. I and II
- C. II and IV
- D. I and III

2. Which learning technique should be used to train an automotive robot painter?

- A. Classification
- B. Regression
- C. Clustering
- D. Reinforcement learning

3. Which of the following is **TRUE** about the roles of data scientists?

- I. They perform data backup.
- II. They decide the appropriate type of visualization tool.
- III. They perform data cleaning.
- IV. They decide the sources of data.

- A. I, II, and IV
- B. I, II and III
- C. I, III and IV
- D. II, III and IV

4. Below is the correct selection of classification algorithm, EXCEPT:
- A. Decision tree
  - B. Random forest
  - C. K-Means algorithm
  - D. K-nearest neighbour

Figure 1.0

|              | Predicted:<br>0 | Predicted:<br>1 |
|--------------|-----------------|-----------------|
| Actual:<br>0 | 8               | 2               |
| Actual:<br>1 | 0               | 1               |

5. What is the correct interpretation of the above confusion matrix?
- A. The value of type I error is 0.
  - B. The value of type II error is 1.
  - C. The recall value is 0.8.
  - D. The precision value is 1.

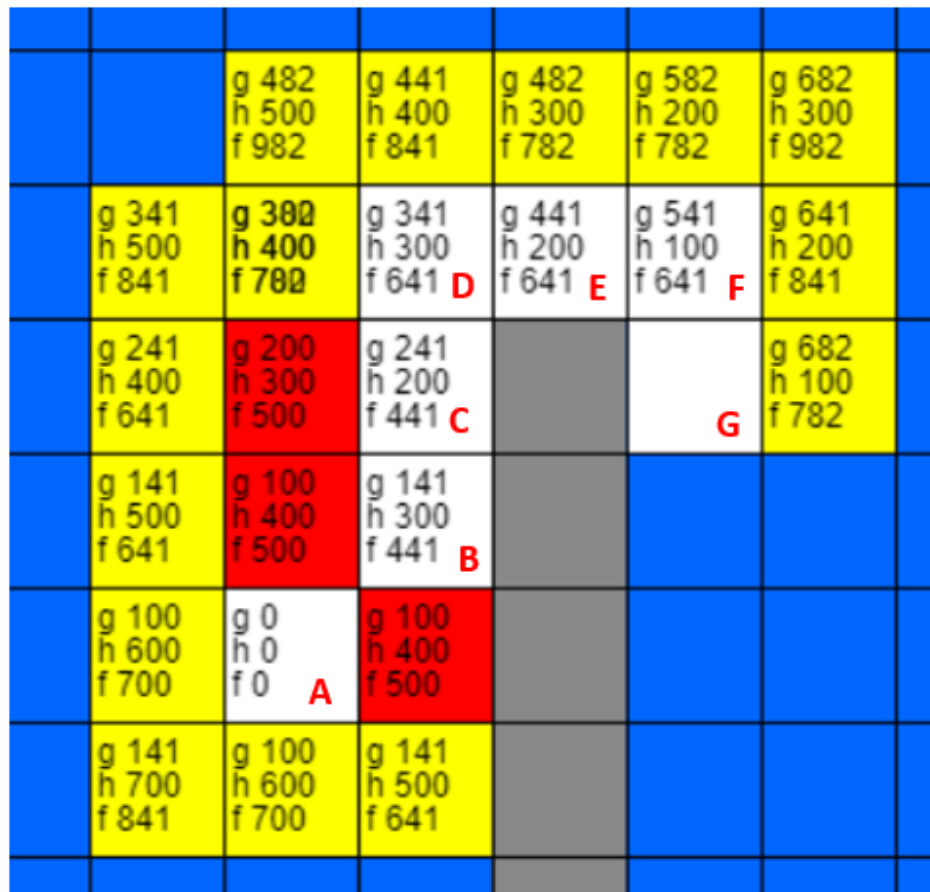
6. This analytic is used to summarize or turn data into relevant information. In other words, it summarized what has occurred. This type of analytics has some meaningful impact but won't be much helpful in forecasting.

The description above is referring to \_\_\_\_\_.

- A. Descriptive analytic
- B. Prescriptive analytic
- C. Diagnostic analytic
- D. Predictive analytic

## SOALAN 1 -CH 4B Search Algorithms

1. Figure 1 below is a maze with tile A as the starting point and tile G as the goal. Using A\* search, the solution path is given as A-B-C-D-E-F-G (white tiles). Yellow tile represent the node in open list and red tile is the node in closed list. Gray tile is the wall where no movement is allowed. Based from the Figure 1, please answer the following questions [5m]



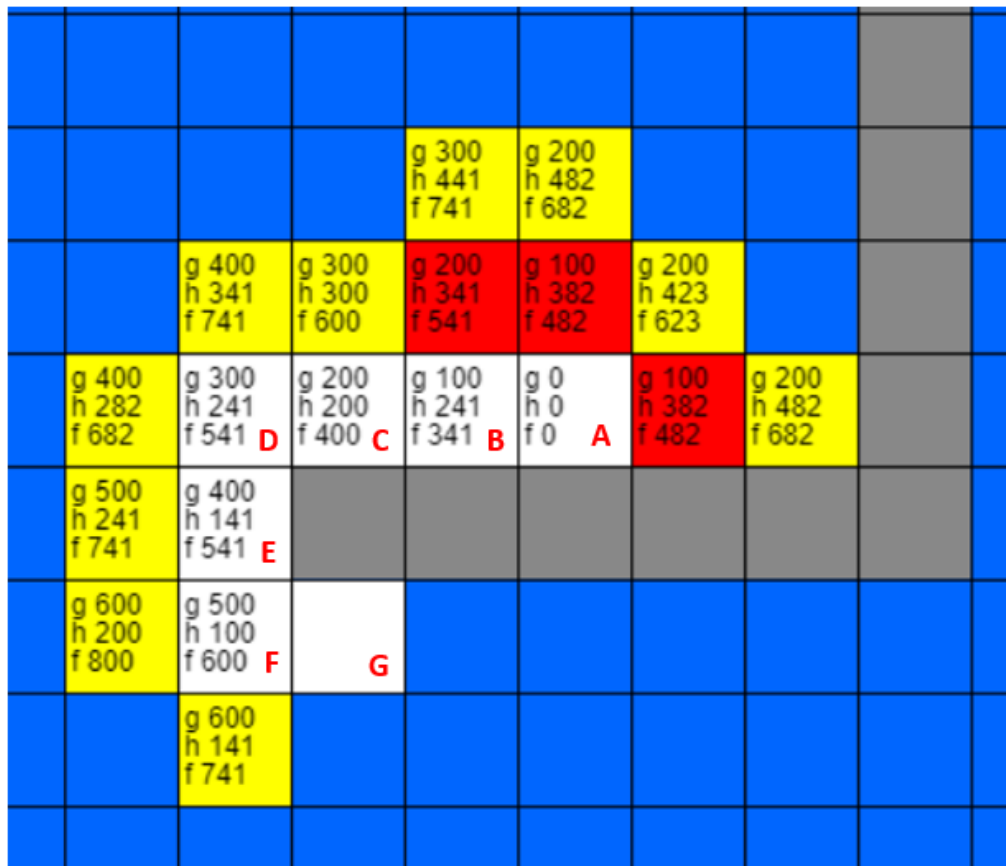
- a) Using  $h(n)$  as Manhattan distance, calculate the solution path by completing the table below [3m]

| Path to state expanded | Manhattan distance $h(n)$ | Cost of n from A, $g(n)$ | $f(n) = h(n) + g(n)$ |
|------------------------|---------------------------|--------------------------|----------------------|
| A                      | 0                         | 0                        | 0                    |
| A-B                    | 300                       | 141                      | 441                  |
| A-B-C                  | 200                       | 241                      | 541                  |
| A-B-C-D                | 300                       | 341                      | 641                  |
| A-B-C-D-E              | 200                       | 441                      | 641                  |
| A-B-C-D-E-F            | 100                       | 541                      | 641                  |
| A-B-C-D-E-F-G          | 0                         | -                        | -                    |

- b) Show that the heuristic is **admissible** or not admissible [2m]

$$h(n) < h^*(n)$$

1. Figure 1 below is a maze with tile A as the starting point and tile G as the goal. Using A\* search, the solution path is given as A-B-C-D-E-F-G (white tiles). Yellow tile represent the node in open list and red tile is the node in closed list. Gray tile is the wall where no movement is allowed. Based from the Figure 1, please answer the following questions [5m]



- a) Using  $h(n)$  as Manhattan distance, calculate the solution path by completing the table below [3m]

| Path to state expanded | Manhattan distance $h(n)$ | Cost of n from A, $g(n)$ | $f(n) = h(n) + g(n)$ |
|------------------------|---------------------------|--------------------------|----------------------|
| A                      | 0                         | 0                        | 0+0                  |
| A-B                    | 241                       | 100                      | 341                  |
| A-B-C                  | 200                       | 200                      | 400                  |
| A-B-C-D                | 241                       | 300                      | 541                  |
| A-B-C-D-E              | 141                       | 400                      | 541                  |
| A-B-C-D-E-F            | 100                       | 500                      | 600                  |
| A-B-C-D-E-F-G          | 0                         |                          |                      |

- b) Show that the heuristic is admissible or not admissible [2m]

1. Let,

A = the ignition key is on

B = the engine will not start

C = headlights work

D = the voltage test on the ignition switch shows 1 to 6 volts

(A, B, C and D are the FACTS in the Initial Database)

E = the starting system (including battery) is faulty

G = starter is faulty

I = battery is dead

F = wiring between the ignition and the solenoid is good

H = replace the ignition switch

(E, F, G, H, I are the FACTS that can be inferred from the FACTS in the Initial Database)

Rules are given below and they are in the order of R1, R2, R3, R4 and R5:

R1:  $A \wedge B \rightarrow E$

R2:  $E \wedge C \rightarrow G$

R3:  $E \wedge \sim C \rightarrow I$

R4:  $D \rightarrow F$

R5:  $F \rightarrow H$

Consider conflict set as a queue and rule must be fired from front of the queue

(a) Use the forward chaining to infer all the possible FACTS from Initial Database (10M)

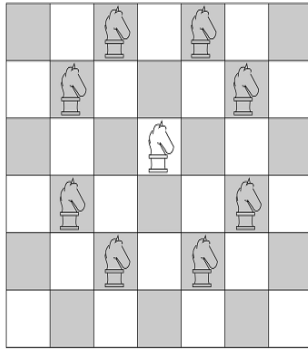
| Iteration # | Working Memory  | Conflict Set | Rule Fired |
|-------------|-----------------|--------------|------------|
| 1           | A, B, C, D      | R1,R4        | R1         |
| 2           | A,B,C,D,E       | R1,R2,R4     | R2         |
| 3           | A,B,C,D,E,G     | R1,R2,R4     | R4         |
| 4           | A,B,C,D,E,G,F   | R1,R2,R4,R5  | R5         |
| 5           | A,B,C,D,E,F,G,H | HALT         | HALT       |

(b) Use backward chaining to infer (H). (Hint: Stop the iteration when sub-goal is able to reach FACT in the Initial Database (either A, B, C or D) (5M)

| Iteration # | Working Memory | Conflict Set | Rule Fired |
|-------------|----------------|--------------|------------|
| 1           | H              | R5           | R5         |
| 2           | H,F            | R4,R5        | R4         |
| 3           | H,F,D          | HALT         | HALT       |

(a) Given a problem of knight's tour in Figure 1 and production rules for the 3 X 3 knight problem in Table 1, please state a production system that shows a knight make a move from 1 to 4. Show your answer as shown in Table 4. (5M)

1.



|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

**Table 1**

| RULE # | CONDITION          | ACTION                    |
|--------|--------------------|---------------------------|
| 1      | knight on square 1 | → Move knight to square 8 |
| 2      | knight on square 1 | → Move knight to square 6 |
| 3      | knight on square 2 | → Move knight to square 9 |
| 4      | knight on square 2 | → Move knight to square 7 |
| 5      | knight on square 3 | → Move knight to square 4 |
| 6      | knight on square 3 | → Move knight to square 8 |
| 7      | knight on square 4 | → Move knight to square 9 |
| 8      | knight on square 4 | → Move knight to square 3 |
| 9      | knight on square 6 | → Move knight to square 1 |
| 10     | knight on square 6 | → Move knight to square 7 |
| 11     | knight on square 7 | → Move knight to square 2 |
| 12     | knight on square 7 | → Move knight to square 6 |
| 13     | knight on square 8 | → Move knight to square 3 |
| 14     | knight on square 8 | → Move knight to square 1 |
| 15     | knight on square 9 | → Move knight to square 2 |
| 16     | knight on square 9 | → Move knight to square 4 |

**Answer:**

| Iteration | Working Memory |      | Conflict Set (Rule No) | Fire Rule |
|-----------|----------------|------|------------------------|-----------|
|           | Current        | Goal |                        |           |
| 0         | 1              | 4    | 1,2                    | 1         |
| 1         | 8              | 4    | 13,14                  | 13        |
| 2         | 3              | 4    | 5,6                    | 5         |
| 3         | 4              | 4    | HALT                   | HALT      |

(b) This is a Water Jug Problem. You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. How can you get exactly 2 gallons of water in the 4-gallon jug and 3 gallons of water in 3-gallon jug? The start state is both jugs are empty.

**Production Rules given are :**

**R1 : Fill 4-gal jug**

**R2 : Fill 3-gal jug**

**R3 : Empty 4-gal jug on ground**

**R4 : Empty 3-gal jug on ground**

**R5 : Pour water from 3-gal jug to fill 4-gal jug**

**R6 : Pour water from 4-gal jug to fill 3-gal jug**

**R7 : Pour all of water from 3-gal jug into 4-gal jug**

**R8 : Pour all of water from 4-gal jug into 3-gal jug**

Use data-drive search approach and fill in the table below :

(Note : Use notation (4-gal Jug, 3-gal Jug) to represent Jug status.

i.e (0,3) means 4-gal jug is empty and 3-gal Jug has full 3 gallons of water) **(10M)**

**ANSWER:**

| Iteration # | Working Memory |            | Conflicts Set<br>(Rule #) | Fired Rule |
|-------------|----------------|------------|---------------------------|------------|
|             | Current State  | Goal State |                           |            |
| 1           | (0,0)          | (2,3)      | 1,2                       | 1          |
| 2           | (4,0)          |            | 2,3,6                     | 6          |
| 3           | (1,3)          |            | 3,4                       | 4          |
| 4           | (1,0)          |            | 2,3,8                     | 8          |
| 5           | (0,1)          |            | 1,4,7                     | 1          |
| 6           | (4,1)          |            | 6                         | 6          |
| 7           | (2,3)          |            |                           |            |

## SET 1

### CHAPTER 6: ADVANCED ARTIFICIAL INTELLIGENCE:

A self-driving car, also known as an autonomous vehicle (AV), driverless car, or robotic car (robo-car), a car incorporating vehicular automation. That is, a ground vehicle which capable of sensing its environment and moving safely with less or without human input.

Table 1: Autonomous Vehicles Agents PEAS

| EXAMPLE            | PERFORMANCE<br>/ GOAL | ENVIRONMENT | SENSOR /<br>PERCEPT | ACTUATOR /<br>ACTION |
|--------------------|-----------------------|-------------|---------------------|----------------------|
| Autonomous vehicle | A                     | B           | C                   | D                    |

1. Referring to Table 1, what is the performance (A) I for the case study? (2 marks)
  - a. Able to stop when detect obstacles
  - b. Accelerate on the normal driving
  - c. Decide the shortest path
2. Referring to Table 1, what is the performances (A) II for the case study? (2 marks)
  - a. Monitor driving environments
  - b. Able to understand the signage
  - c. Use the traffic information and update current traffic
3. Referring to Table 2, what is the Environment (B) for the case study? (2 marks)
  - a. Road or Highway
  - b. Person who are in charge to monitor AV car
  - c. Weather condition
4. Referring to Table 1, what is the Sensor (C) for the case study? (2 marks)
  - a. GPS
  - b. Breaking system
  - c. Traffic light
5. Referring to Table 1, what are the Actuator (D) for the case study? (2 marks)
  - a. Making decision based on the environment to manoeuvre through the traffic
  - b. Update information and find the shortest path
  - c. Able to track the driver's location

Alexa is Amazon's virtual assistant technology. Alexa can help on the simple task such as play favourite song, read the latest headlines, dim the lights in living room, and more.

Table 2: Alexa Agents PEAS

| EXAMPLE | PERFORMANCE<br>/ GOAL | ENVIRONMENT | SENSOR /<br>PERCEPT | ACTUATOR /<br>ACTION |
|---------|-----------------------|-------------|---------------------|----------------------|
| Alexa   | A                     | B           | C                   | D                    |

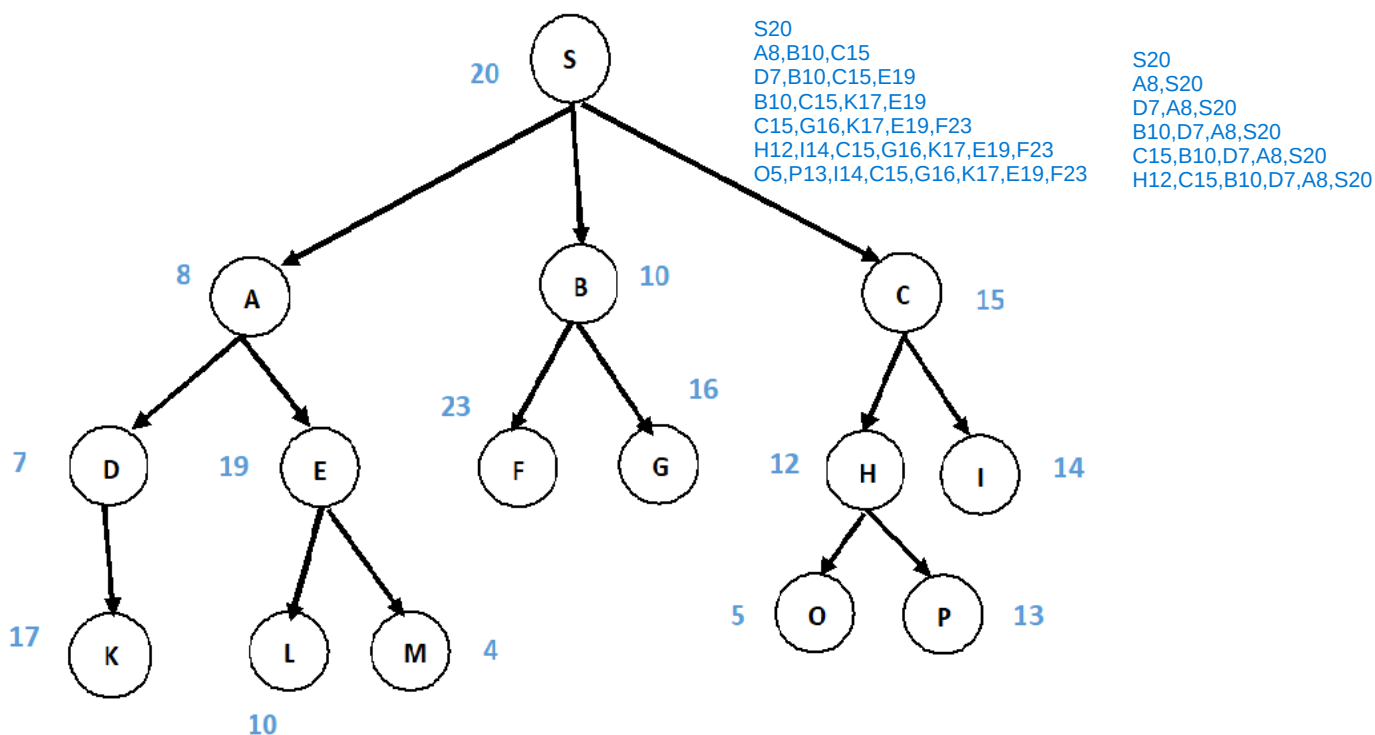
1. Referring to Table 2, what are the performance (A) I for the case study? (2 marks)
  - a. Able to record voice
  - b. Able to write information
  - c. Respond correctly to the user query
2. Referring to Table 2, what are the performances (A) II for the case study? (2 marks)
  - a. Able to detect and tracking user location
  - b. Upload new information based on user request
  - c. Check weather forecasting regularly

3. Referring to Table 2, what is the Environment (B) for the case study? (2 marks)
  - a. Speech recognition
  - b. User Information
  - c. Weather information
4. Referring to Table 2, what is the Sensors (C) for the case study? (2 marks)
  - a. Voice
  - b. GPS Tracker
  - c. Camera
5. Referring to Table 2, what are the Actuator(D) for the case study? (2 marks)
  - a. Using sensors to perceive user request and automatically collect data from the internet
  - b. Update information and manage the user social account
  - c. Manage the information and track the user locations

#### QUESTION 4 -CH 4A Search Algorithms

##### SET 1 - 15 M

a. Consider the following graph. Find the most cost-effective path from initial state S to reach the goal state O using Best First Search Algorithm.

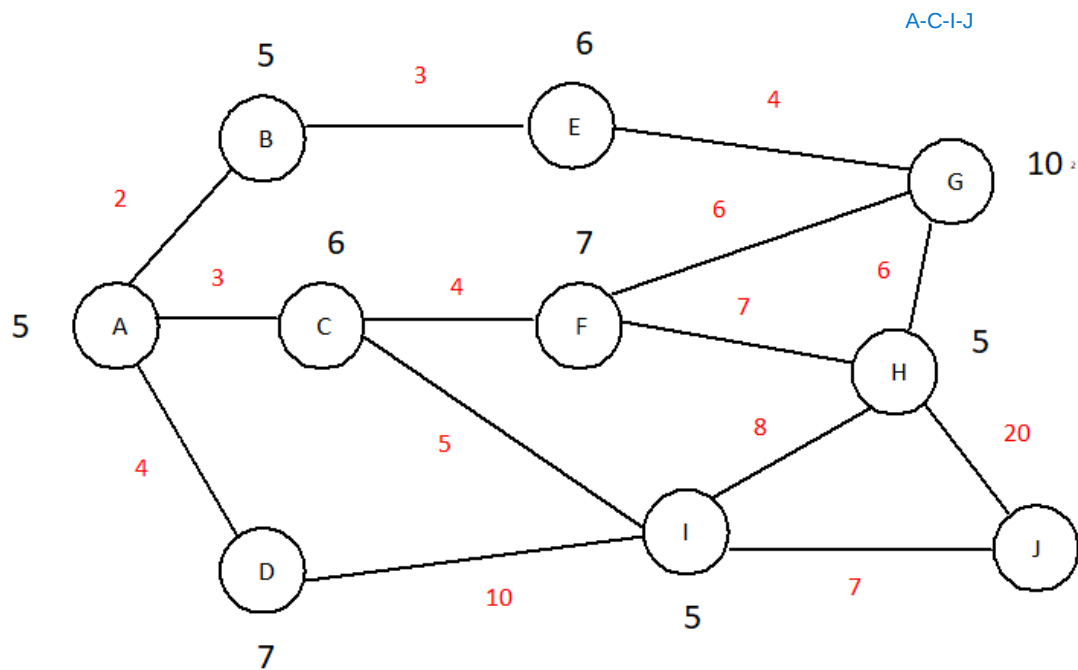


**Answer (5 marks)**

We start from source "S" and search for goal "O" using given costs and Best First search.

| Node | g(n) | h(n) | f(n) |
|------|------|------|------|
| A    | 0    | 5    | 5    |
| B    | 2    | 5    | 7    |
| C    | 3    | 6    | 9    |
| D    | 4    | 7    | 11   |
| E    | 5    | 6    | 11   |
| F    | 7    | 7    | 14   |
| G    | 9    | 10   | 19   |
| H    | 14   | 5    | 19   |
| I    | 8    | 5    | 13   |
| J    | 15   | 0    | 15   |

b. Consider the following directed graph.



The numbers written on edges represent the distance between the nodes.

The numbers written on nodes represent the heuristic value.

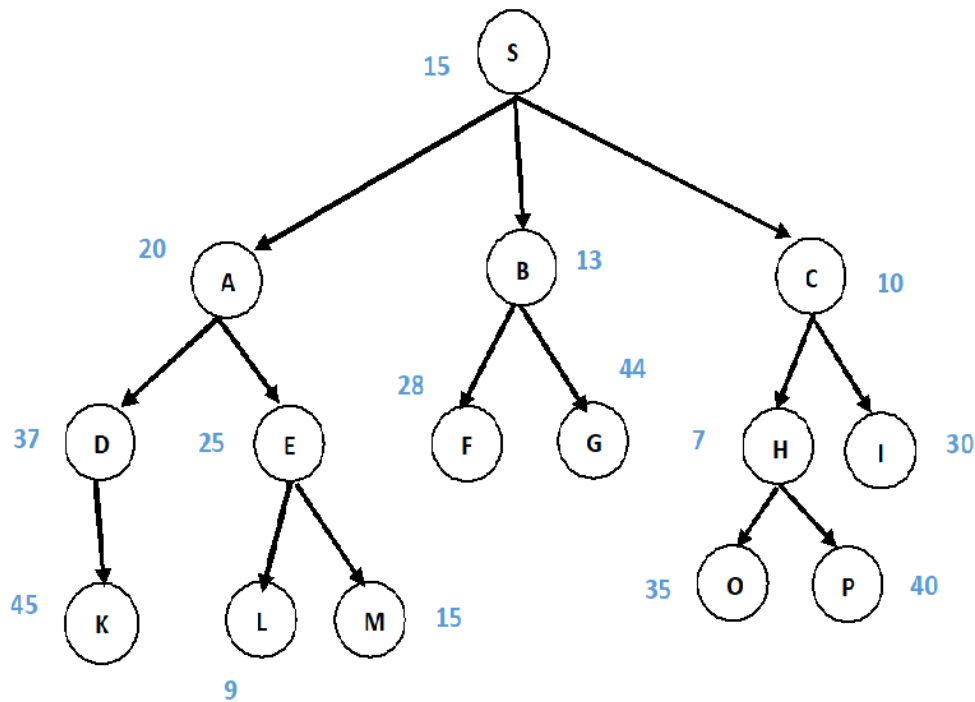
Find the most cost-effective path to reach from start state A to final state J using A\* Algorithm. (10 M)

SET 2 – 15 M

a. Consider the following graph. Find the most cost-effective path from initial state S to reach the goal state O using Best First Search Algorithm. (5M)

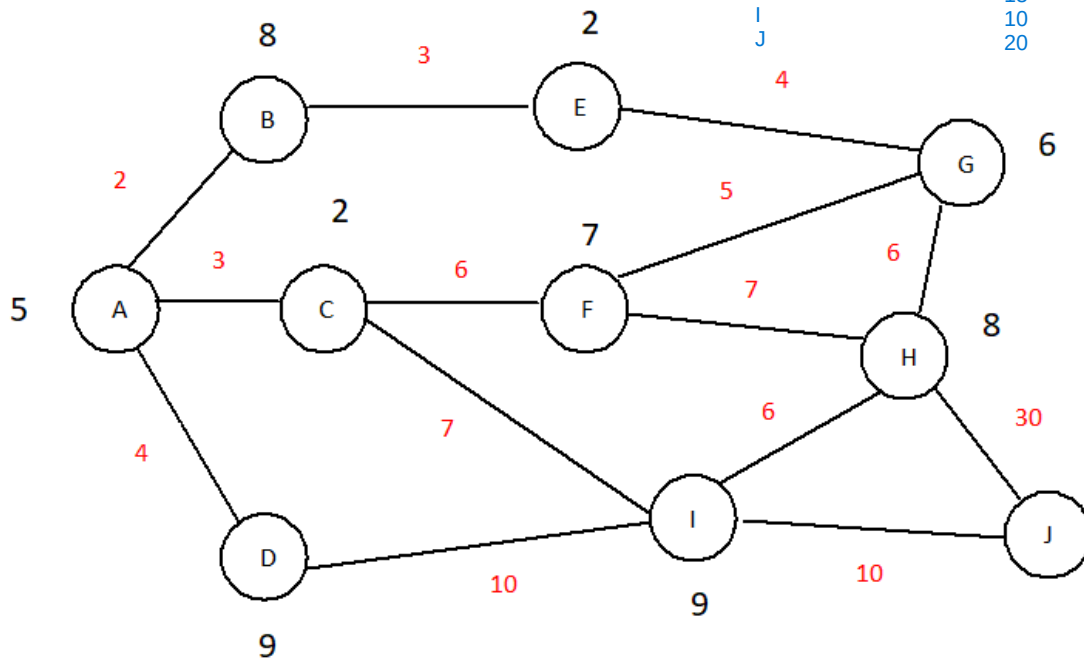
S15  
 C10,B13,A20  
 H7,B13,A20,I30  
 B13,A20,I30,O35,P40  
 A20,F28,I30,O35,P40,G44  
 E25,F28,I30,O35,D37,P40,G44  
 L9,M15,F28,I30,O35,D37,P40,G44

S15  
 C10,S15  
 H7,C10,S15  
 B13,H7,C10,S15  
 A20,B13,H7,C10,S15  
 E25,A20,B13,H7,C10,S15



We start from source "S" and search for goal "L" using given costs and Best First search.

b. Consider the following directed graph. (10 m)



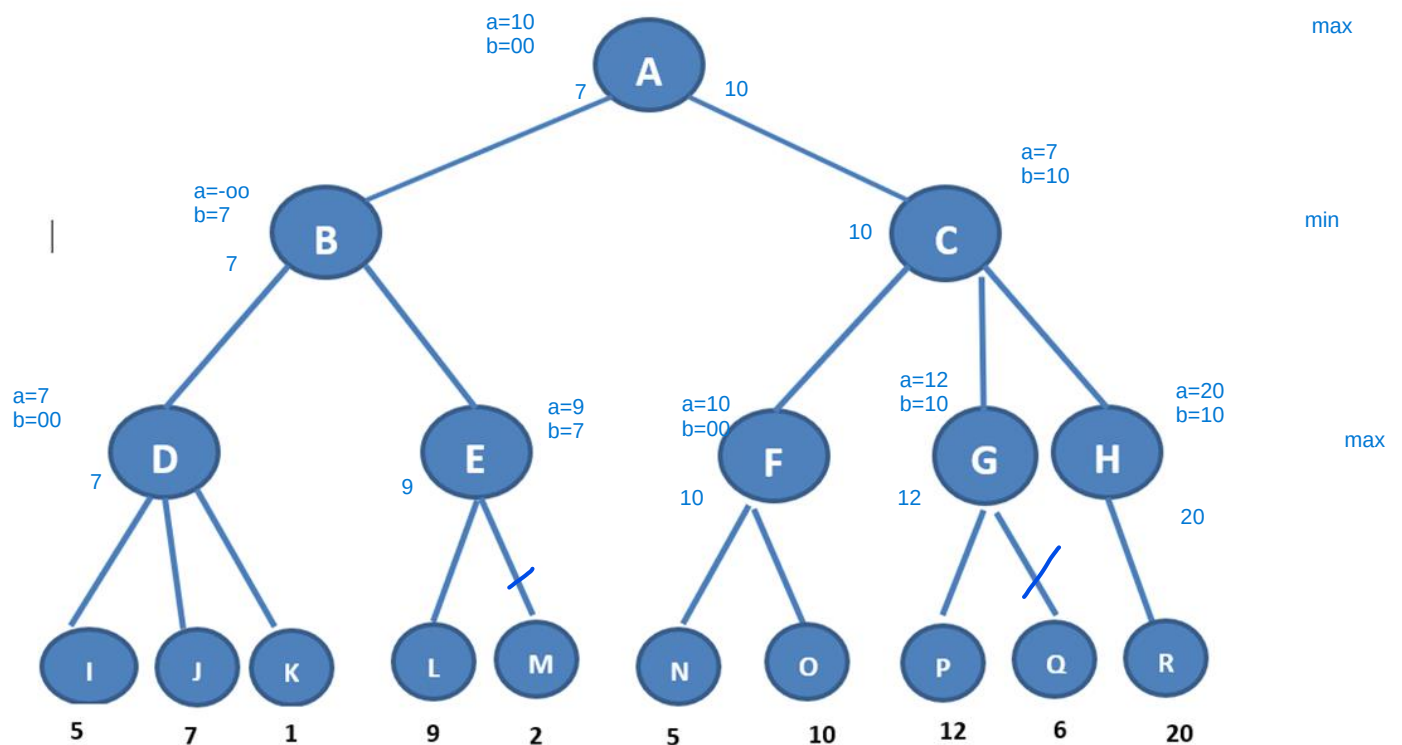
| Node | g(n) | h(n) | f(n) |
|------|------|------|------|
| A    | 0    | 5    | 5    |
| B    | 2    | 8    | 10   |
| C    | 3    | 2    | 5    |
| D    | 4    | 9    | 13   |
| E    | 5    | 2    | 7    |
| F    | 9    | 7    | 16   |
| G    | 9    | 6    | 15   |
| H    | 15   | 8    | 23   |
| I    | 10   | 9    | 19   |
| J    | 20   | 0    | 20   |

A-C-I-J

The numbers written on edges represent the distance between the nodes.  
The numbers written on nodes represent the heuristic value.

Find the most cost-effective path to reach from start state A to final state J using A\* Algorithm.

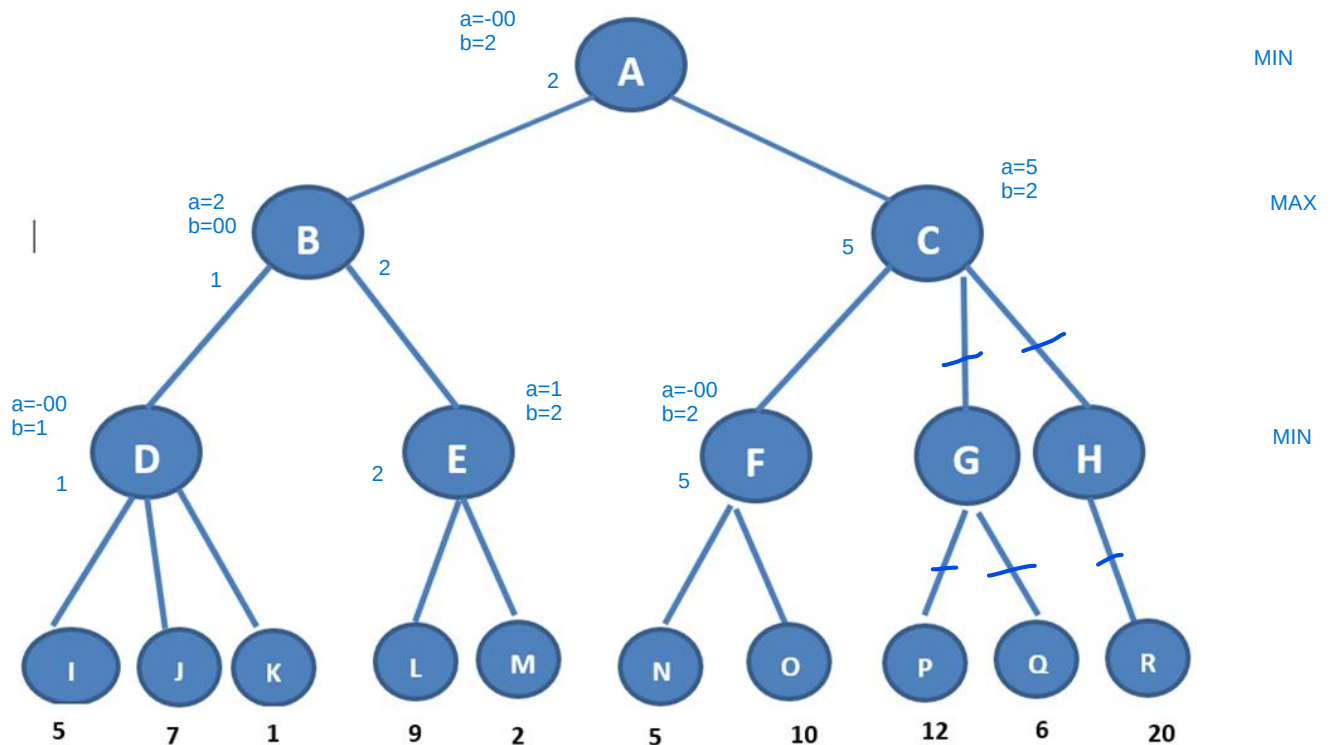
### QUESTION 5 CH 4C Problem Solving with search



**Note : MAX start first**

You are required to apply Minimax with Alpha Beta pruning procedure to the search tree in the figure above. Analyze and calculate the values for the following questions. [10m]

- Calculate the values for nodes A , B, C, D, E, F, G and H? 10,7,10,7,9,10,12,20
- Calculate the  $\alpha$  and  $\beta$  final values for node C ? 7,10
- Examine and select the nodes that will not be evaluated. M,Q
- Calculate the optimal value for MAX to win the game? 10
- Examine and select all the nodes in the optimal path for MAX to win the game. A-C-F-O

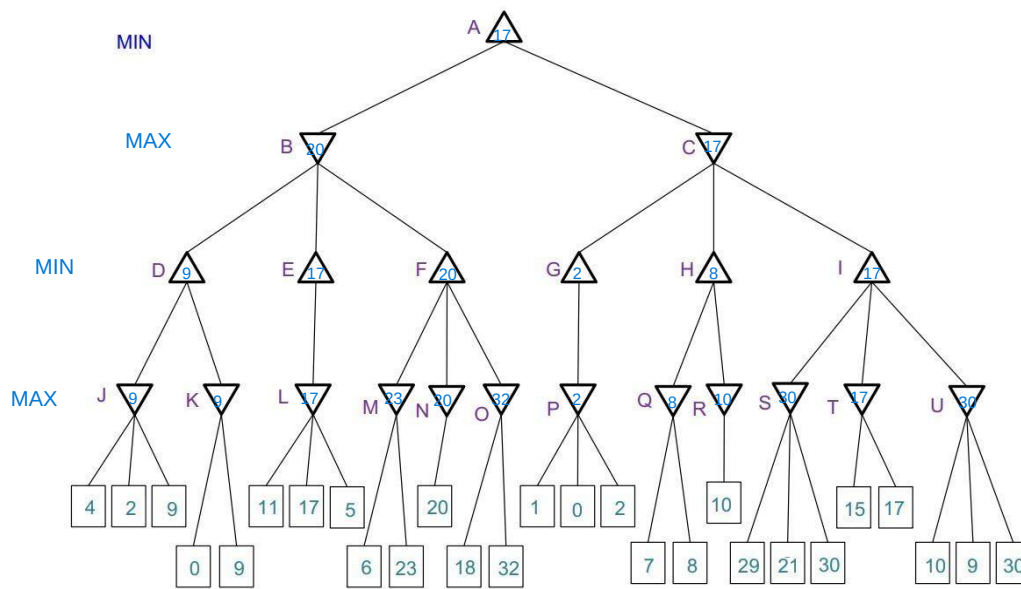


**Note : MIN start first**

You are required to apply Minimax with Alpha Beta pruning procedure to the search tree in the figure above. Analyze and calculate the values for the following questions. [10m]

- Calculate the values for nodes A , B, C, D, and E ? 2,2,5,1,2
- Calculate the  $\alpha$  and  $\beta$  final values for node C ? 5,2
- Examine and select the nodes that will not be evaluated. G,H,P,Q,R
- Calculate the optimal value for MIN to win the game? 2
- Examine and select all the nodes in the optimal path for MIN to win the game. A-B-E-M

Complete all states (A – U) in the following tree using MINIMAX without Alpha Beta pruning. (10 m)



## QUESTION 7

Figure A shows COVID-19 attributes and their classification results (assume COVID-19 positive case be 1 and the negative case is 0). Below are the detailed attributes.

Attributes:

1. LC - Location
2. CO- Country
3. GE – Gender
- 4- AG- Age
- 5- CH-Cough
- 6- CP- Chest Pain
- 7- FE- Fever

| LC  | CO | GE | AG | CH | CP | FE | TESTED_POSITIVE | PREDICTED_POSITIVE |
|-----|----|----|----|----|----|----|-----------------|--------------------|
| 104 | 8  | 1  | 66 | 1  | 0  | 1  | 1               | 0                  |
| 101 | 7  | 0  | 56 | 0  | 1  | 0  | 1               | 0                  |
| 137 | 6  | 1  | 46 | 0  | 1  | 0  | 1               | 1                  |
| 116 | 9  | 0  | 60 | 1  | 0  | 0  | 0               | 0                  |
| 116 | 6  | 1  | 58 | 0  | 0  | 0  | 0               | 0                  |
| 23  | 6  | 0  | 44 | 0  | 1  | 0  | 1               | 1                  |
| 105 | 7  | 1  | 34 | 0  | 1  | 0  | 1               | 1                  |
| 13  | 7  | 1  | 37 | 1  | 0  | 0  | 0               | 1                  |
| 13  | 8  | 1  | 39 | 1  | 0  | 0  | 1               | 0                  |
| 13  | 8  | 1  | 56 | 1  | 0  | 0  | 1               | 0                  |
| 13  | 9  | 0  | 18 | 1  | 0  | 0  | 1               | 1                  |
| 13  | 9  | 0  | 32 | 1  | 0  | 0  | 0               | 1                  |
| 100 | 7  | 1  | 37 | 1  | 0  | 0  | 0               | 1                  |
| 135 | 7  | 1  | 51 | 0  | 1  | 0  | 0               | 0                  |
| 105 | 7  | 1  | 57 | 0  | 1  | 1  | 1               | 1                  |
| 53  | 6  | 1  | 56 | 1  | 0  | 0  | 0               | 0                  |
| 53  | 6  | 1  | 50 | 1  | 0  | 0  | 0               | 0                  |
| 71  | 8  | 0  | 52 | 0  | 1  | 0  | 0               | 0                  |
| 67  | 9  | 1  | 33 | 1  | 0  | 0  | 0               | 1                  |
| 67  | 8  | 1  | 40 | 1  | 0  | 0  | 1               | 1                  |

Based on the dataset given in Figure A, answer the following questions:

a) Quantify the amount of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) for the classification result

TP - 6  
TN - 6  
FP - 4  
FN - 4

[4 m]

b) Using your answers in a), calculate the accuracy, precision, and recall for this classification

Accuracy - 0.6  
Precision - 0.6  
Recall - 0.6

[6 m]

Figure A shows Neighbourhood X attributes in urban areas and their binary classification results. Below are the detailed attributes.

| Household | Gender | Car_type | Property | Class_T20 | Predicted_classT20 |
|-----------|--------|----------|----------|-----------|--------------------|
| 1         | M      | Family   | Semi_d   | T20       | T20                |
| 2         | M      | Sports   | Bungalow | M40       | M40                |
| 3         | M      | Sports   | Semi_d   | M40       | M40                |
| 4         | M      | Sports   | Semi_d   | T20       | T20                |
| 5         | M      | Sports   | Semi_d   | M40       | T20                |
| 6         | M      | Luxury   | Semi_d   | M40       | T20                |
| 7         | F      | Luxury   | Semi_d   | M40       | M40                |
| 8         | F      | Luxury   | Terrace  | T20       | T20                |
| 9         | F      | Luxury   | Bungalow | M40       | M40                |
| 10        | F      | Luxury   | Terrace  | M40       | M40                |
| 11        | M      | Luxury   | Semi_d   | M40       | M40                |
| 12        | M      | Family   | Bungalow | T20       | T20                |
| 13        | M      | Family   | Bungalow | T20       | T20                |
| 14        | M      | Sports   | Bungalow | M40       | M40                |
| 15        | F      | Family   | Terrace  | T20       | T20                |
| 16        | F      | Sports   | Semi_d   | T20       | M40                |
| 17        | F      | Sports   | Terrace  | T20       | M40                |
| 18        | F      | Family   | Terrace  | T20       | M40                |
| 19        | F      | Family   | Terrace  | T20       | T20                |
| 20        | F      | Luxury   | Semi_d   | M40       | M40                |

Based on the dataset given in Figure A, answer the following questions:

a) Quantify the amount of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) for the classification result. **[4 m]**

TP - 7  
TN - 8  
FP - 2  
FN - 3

b) Using your answers in a), calculate the accuracy, precision, and recall for this classification

Accuracy - 0.75  
Precision - 0.78  
Recall - 0.7

**[6 m]**